

Engineering Handbook – Volume 06

Telemetry & Industrial IoT Architecture (Version 1.0)

Purpose

Defines the end-to-end telemetry architecture from industrial equipment to cloud dashboards.

Industrial Devices

Initial deployment targets INVT VFDs, Jet Dyeing Machines, Stenters, Pumps, Blowers, Thermic Fluid Heater, Boilers and utility meters.

Edge Gateway

A Raspberry Pi gateway communicates with Modbus RTU devices through RS485, timestamps readings, buffers outages and forwards validated payloads.

Communication

Modbus RTU is the primary field protocol. MQTT is the planned message transport. HTTPS REST APIs remain available for provisioning and administration.

Telemetry Pipeline

Sensor → PLC/VFD → RS485 → Raspberry Pi → Parser → Validation → FastAPI → PostgreSQL → Analytics → Dashboards → Mobile Notifications.

Data Model

Every reading belongs to Company → Machine → Component → Tag. Readings contain timestamp, value, quality flag and source metadata.

Polling Strategy

Different tags use configurable polling intervals. Critical electrical parameters may be collected every second while slower process variables use longer intervals.

Quality & Reliability

Gateway buffering, retries, communication status, stale data detection and quality flags ensure reliable industrial data acquisition.

Alarm Integration

Telemetry feeds threshold alarms, rate-of-change alarms, communication-loss alarms and predictive maintenance models.

Future Roadmap

MQTT broker, Redis cache, stream processing, anomaly detection, historian optimization, compression and AI-assisted diagnostics.

Reference Data Flow

1. Poll device registers
2. Decode engineering values
3. Validate payload
4. Buffer if offline
5. Publish to backend
6. Persist to database
7. Aggregate KPIs
8. Trigger alarms
9. Display dashboards
10. Notify users